1. I take 10 cards from a well-shuffled standard deck of 52 cards.

   (a) What is the probability that I will get exactly 2 aces?

   (b) What is the probability that I will get 3 aces and 2 kings?

2. Simplify
$$\binom{5}{0}\binom{6}{2}+\binom{5}{1}\binom{6}{3}+\binom{5}{2}\binom{6}{4}+\binom{5}{3}\binom{6}{5}+\binom{5}{4}\binom{6}{6}.$$

3. Solve the recurrence equation
$$a_n - 4a_{n-2} = 1, \quad a_0 = 0, a_1 = 1.$$

4. You play a the following lottery with the casino. There is a hat on the table. A fair dice is thrown until the game ends.

   - If the dice shows one then $+1$ dollar is added to the hat and the game continues.
   - If the dice shows two then you immediately receive $+2$ dollars and the game continues.
   - If the dice shows three then $+3$ dollars are added to the hat and the game continues.
   - If the dice shows four then your receive half of the sum in the hat and the game ends.
   - If the dice shows five then the sum in the hat goes back to the casino and the game continues.
   - If the dice shows six then you receive the content of the hat and the game ends.

   Let e_n be your total expected payoff if there are n dollars in the hat now.

   Find the recurrence equation for e_n. You don't need to solve it.

5. Consider the recurrence equation
$$a_n - 5a_{n-1} + 3a_{n-2} = 0 \text{ for } n \geq 2, \quad a_0 = 1, a_1 = 2.$$

   Find the generating function $g(t)$ for the sequence $(a_0, a_1, a_2, \dots)$.

6. The sequence $(b_0, b_1, b_2, \dots)$ has a generating function
$$g(t) = \frac{3}{1-3t} + \frac{2t}{1-2t}.$$

   Find b_0, b_1 and b_n.